

BraTS Glioma Segmentation Project

Group Members: Please fill Chinese names here

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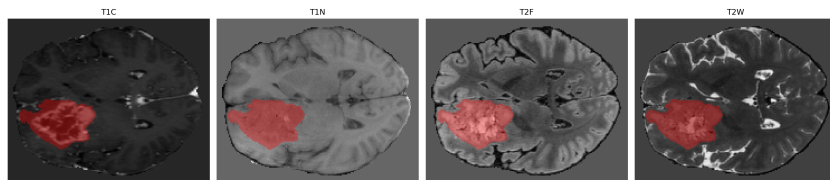
Project Goal

- Build an automatic glioma segmentation pipeline on BraTS MRI data.
- Cover preprocessing/visualization, baseline training, and advanced-model analysis.
- Run everything locally with ROCm PyTorch in `/home/tzh03/myenv`.

Dataset and Preprocessing

- Local BraTS 2023 dataset: 560 verified cases, four modalities plus segmentation.
- Verified volume size: $240 \times 240 \times 155$.
- Implemented z-score normalization, foreground cropping, and WT/TC/ET conversion.
- Exported positive 2D slices for training.

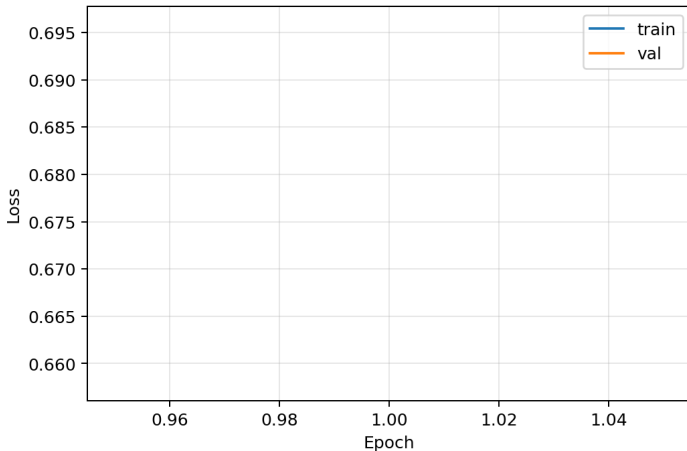
Task 1 Result



Best ET contrast modality in the executed analysis: **T1C**

- Baseline: 2D U-Net.
- Upgrade: Attention U-Net.
- Loss: Dice + BCE.
- Augmentation: flips, rotations, scaling, Gaussian noise.
- Metrics: Dice for WT/TC/ET and HD95.

Baseline Training Artifact



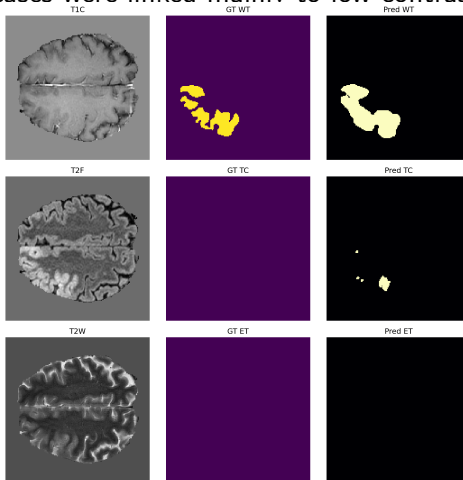
Experiment Comparison

Region	Baseline	Attention	Δ
WT	0.7063	0.6506	-0.0557
TC	0.5409	0.3834	-0.1574
ET	0.8008	0.6560	-0.1448

In this verified smoke experiment, the baseline U-Net outperformed the attention variant.

Worst-case Analysis

- Hardest case in both models: BraTS-GLI-00009-001.
- Dominant failure reason: **small lesion size**.
- Other weak cases were linked mainly to low contrast.



Conclusion

- Delivered a complete local BraTS coursework repository.
- Verified preprocessing, training, evaluation, and error-analysis scripts with real runs.
- The current outputs are smoke-scale but reproducible and ready for longer full training.